

Modelling Learner Behaviour Over Time: Engagement, Self-Regulation and At-Risk Prediction in OULAD and Moodle

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Code & notebooks: https://github.com/hariharateja/ET610_ASSIGNMENT2

Abstract. We analyse two learning-analytics datasets. One is the Open University Learning Analytics Dataset (OULAD; 29,496 enrolments, 10.6 M daily click records). The other is a 146,048-event Moodle log from a blended IT course (161 learners). For each dataset we test hypotheses about early engagement, regularity, self-monitoring and procrastination. We build weekly behavioural trajectories and evaluate supervised and unsupervised models. In OULAD, a Random Forest trained on 2013 cohorts flags at-risk learners in the 2014 cohorts with ROC-AUC 0.76 at week 4, 0.81 at week 8 and 0.93 at week 26. Regularity features add up to +0.07 AUC over click volume. Submission latency, however, has no effect once score and submission rate are controlled for. In Moodle, behaviour from the first ten weeks predicts completion of the final project (AUC 0.94). The strongest signals are early Synopsis submission and self-monitoring of grades and feedback, not raw activity. K-Means separates four profiles whose completion rates range from 8% to 86%.

1. Introduction & Hypotheses

Learning-management-system logs record what learners do and when they do it. Self-regulated learning theory (Zimmerman, 2002) holds that successful learners plan, monitor and adjust their study. Procrastination research (Steel, 2007) links delay to worse outcomes. Both theories predict that *how* and *when* a learner engages matters, not only *how much*. We ask three questions. **RQ1:** How early do the trajectories of successful and unsuccessful learners diverge? **RQ2:** Do timing behaviours (submission latency, regularity, self-monitoring) carry information beyond activity volume? **RQ3:** Can learners be segmented into interpretable behavioural profiles that differ in outcome?

OULAD hypotheses.

O-H1 (early at-risk divergence). Learners who later fail or withdraw already show lower and declining VLE engagement in the first weeks. Early clickstream features alone reach $AUC \geq 0.75$ by week 4.

O-H2 (submission latency). Later submission relative to the deadline is associated with failing or withdrawing, even after controlling for assessment score and submission rate.

O-H3 (regularity over volume). Engagement regularity (active-day and active-week ratios, weekly burstiness, recency, activity diversity) adds predictive value beyond total clicks and demographics.

Moodle hypotheses.

M-H1 (distributed, self-monitored engagement). In weeks 0–9, learners who go on to submit the Final project log in on more distinct days and weeks, have more regular gaps between sessions, and check their own grades and feedback more often.

M-H2 (procrastination & self-testing). Learners who submit the first milestone (Synopsis) close to its deadline, or not at all, are less likely to complete the Final project. Completers also use practice quizzes more. (The deadline is inferred from the data, so we test how early learners submit, not whether they submit late.)

M-H3 (behavioural profiles). Unsupervised clustering yields distinct, interpretable profiles (e.g. steady self-regulators vs. exam-window-only users) with different completion rates.

2. Datasets & Preprocessing

2.1 OULAD

OULAD (Kuzilek et al., 2017) covers 22 presentations of 7 modules (2013–2014). It contains demographics, registration dates, assessment deadlines and submissions, and daily click counts per VLE resource. Cleaning steps: (i) we de-duplicated enrolments on (module, presentation, student). (ii) We removed 3,097 enrolments that unregistered on or before day 0, because they never had a chance to engage. This leaves 29,496 enrolments (Pass 12,361; Fail 7,044; Withdrawn 7,067; Distinction 3,024). (iii) We joined click records to activity types and aggregated them to one row per enrolment and day. (iv) We excluded exam rows from the assessment features, since exam dates are often missing. Banked (carried-over) results were also dropped. (v) Missing IMD bands and registration dates were imputed with the median. The binary target is **at-risk = Fail or Withdrawn**. To mimic a real early-warning system, every feature at cut-off day c uses only events dated before c . Learners who had already withdrawn by c are not

predicted at that cut-off.

2.2 Moodle interaction log

The Moodle export has 146,048 events for course IT-2507 (23 Oct 2025 – 29 Jul 2026). Each event records a user, a timestamp, an event context, a component, a content code and a description. **Clean-up:** (i) for 3,334 SCORM events the *User ID* column was truncated (e.g. `e_id '3`). We recovered the true id from the description text with a regular expression. (ii) We parsed the dd/mm/yy timestamps and checked for exact duplicates (none found). (iii) We dropped 11 accounts with fewer than 30 events, leaving 161 learners. (iv) We excluded *Quiz attempt auto-saved* events (37k rows) from action counts, because they are system-generated during quiz attempts. They are kept for session timing. **Sessionisation:** events from the same user less than 30 minutes apart belong to one session (Kovanović et al., 2015). This gives 6,348 sessions (median 4 events, 1.5 min). A single-event session is credited with 1 minute of time on task. **Outcome & leakage control:** the log contains no grades. Our outcome is therefore **submission of the Final course-project assignment** (58 of 161 learners, 36%). Predictive features use only events before 29 Dec 2025, the day of the first Final submission. That gives an early window of weeks 0–9, from which all Final-assignment events are removed. The Synopsis deadline is inferred as end of day 30 Nov 2025, the date of the last Synopsis submission. Because of this, no submission can be late by construction (the closest one came 0.2 days before the proxy deadline).

3. Feature Extraction & Engineering

Construct	OULAD features (computed before cut-off c)	Moodle features (weeks 0–9)
Volume	log total clicks; log pre-start clicks	log actions; log content views (book, page, SCORM, video, PPT...)
Regularity / time management	active days; active-day ratio (= active days / c); active-week ratio; weekly coefficient of variation (burstiness); days since last login; recent trend (log clicks of last 2 weeks vs. earlier)	sessions; active days; active-week ratio; weekly CV; SD of inter-session gaps; night (00–06 h) and weekend share
Time on task	—	total sessionised hours; mean session length (min)
Breadth / strategy	Shannon entropy of clicks across 20 activity types	distinct items accessed; practice-quiz attempts & share of actions; forum/glossary views
Submission behaviour	assessments due; submission rate; mean latency (days submitted – deadline); mean score	Synopsis submitted (0/1); Synopsis lead time (days before deadline; –5 if not submitted)
Self-monitoring	—	views of own grade report, assignment feedback and submission status
Performance / context	prior attempts, credits, IMD band, education, age, gender, disability, registration day	mean SCORM self-check score

Table 1. Engineered features, grouped by behavioural construct. Clustering also uses an *exam-window share* feature: the share of a learner's actions in weeks 14–15 (26 Jan – 8 Feb 2026, when the continuous-evaluation quizzes ran).

4. Temporal Behavioural Visualisations

4.1 OULAD

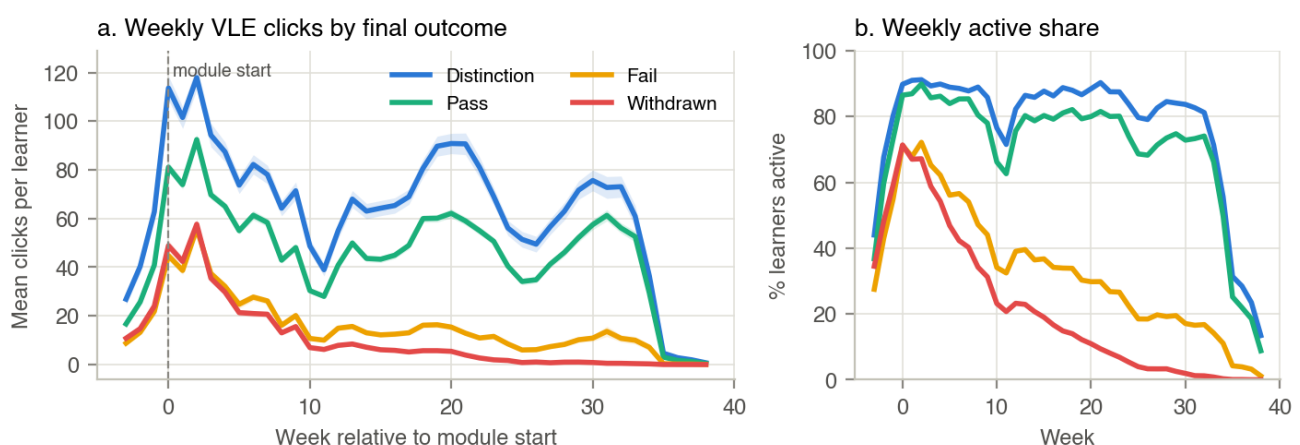


Figure 1. (a) Mean weekly VLE clicks per learner by final outcome (bands = 95% CI). (b) Share of learners active in each week. Week 0 is the module start.

All four groups engage strongly in the first fortnight, but the curves separate almost immediately (Fig. 1). In week 0 the median at-risk learner logs 16 clicks, against 49 for successful learners (rank-biserial $r = 0.30$). By week 4 the medians are 5 vs. 38 ($r = 0.40$). By week 6 the median at-risk learner is no longer active at all ($r = 0.47$). Two patterns stand out. Failing learners do not drop out: their activity stays low and gently declining. Withdrawn learners decay steadily. Successful learners follow a periodic rhythm, with peaks around weeks 2, 20 and 30 (assessment dates differ between modules, so these are pooled peaks). Both Distinction and Pass learners dip in week 11, which looks like a scheduled

break rather than disengagement.

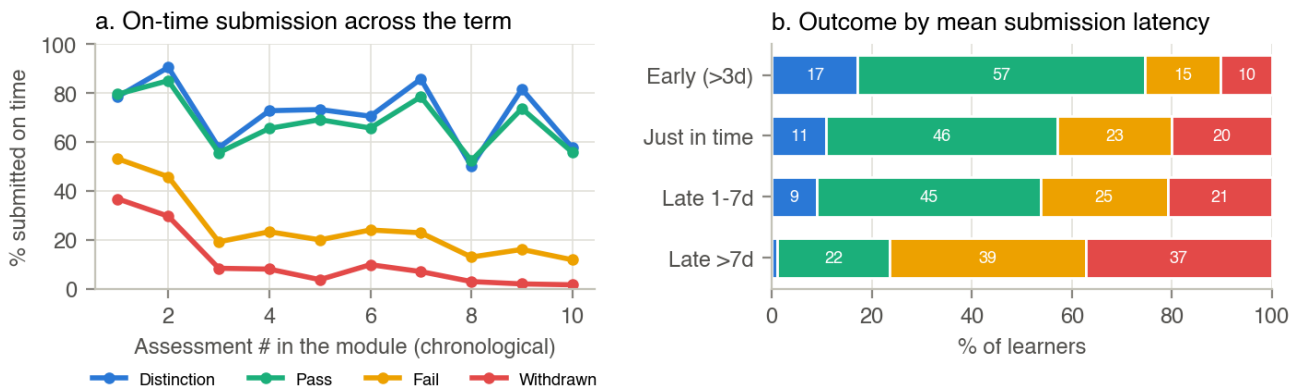


Figure 2. (a) Share of each assessment submitted on or before its deadline, in chronological order within the module. (b) Final-outcome mix by a learner's mean submission latency (submitted assessments only).

Submission punctuality (Fig. 2a) shows the same early separation. On the first assessment, at-risk groups are already 26–42 points behind. By the third assessment fewer than 20% of their assessments are submitted on time, partly because many are never submitted. Mean latency is clearly graded with outcome (Fig. 2b): 25% of learners who submit more than 3 days early are at risk, against 76% of those who are on average more than a week late (Spearman $\rho = 0.24$).

4.2 Moodle

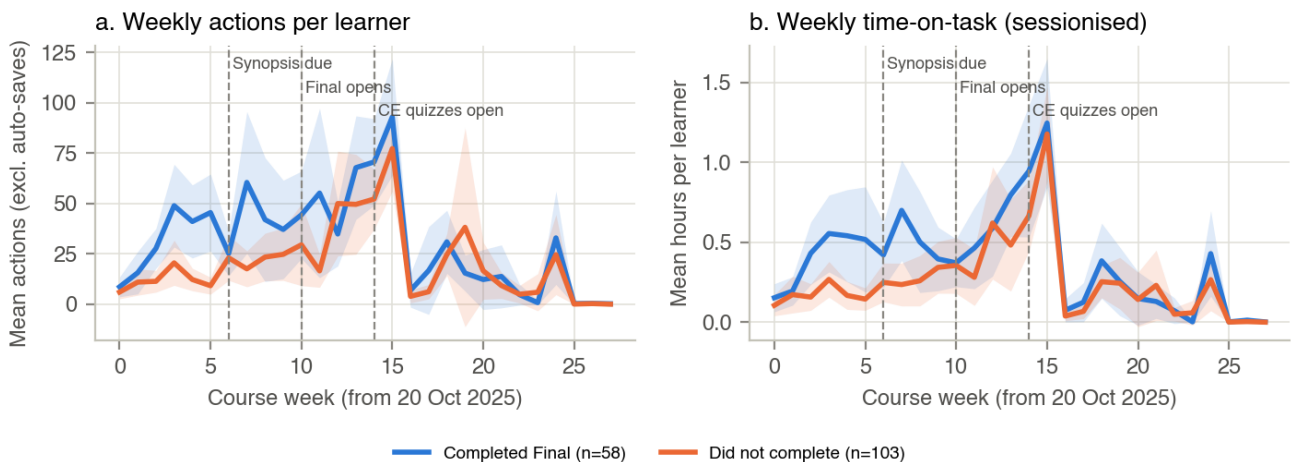


Figure 3. Weekly actions (a) and sessionised time-on-task (b) for learners who did and did not complete the Final project (bands = 95% CI). Dashed lines mark the Synopsis deadline, the opening of Final submissions and the continuous-evaluation (CE) quiz window.

Completers are about twice as active as non-completers through the first ten weeks (Fig. 3). In weeks 2–5 they spend roughly 0.5 h per week on the platform, against about 0.2 h. About a week after the Synopsis deadline (week 7), completers produce a second burst of activity. The gap then closes. In the CE quiz window (weeks 14–15) both groups converge on the same exam-driven peak, and after week 16 their trajectories are indistinguishable. In short, the groups differ in when they engage, not in their exam-time effort: non-completers engage mainly when assessment forces them to.

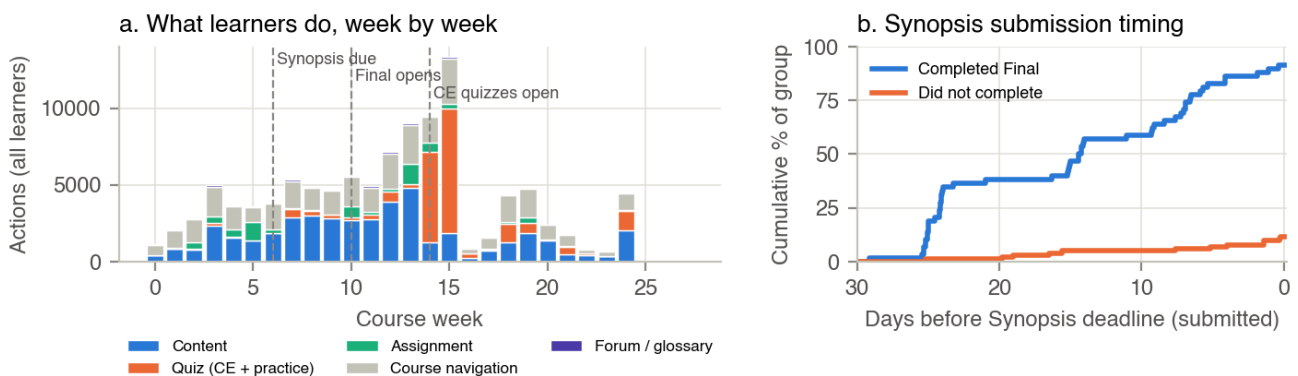


Figure 4. (a) Weekly composition of all learner actions by component (auto-saves excluded). (b) Cumulative share of each group that had submitted the Synopsis, by days before its deadline (read left to right as time passes).

Figure 4a shows how activity moves through the course: content reading in weeks 3–13, assignment work around each milestone, and a sharp switch to quizzes in weeks 14–15. Figure 4b shows the strongest temporal signal. About a third (33%) of eventual completers submitted the Synopsis more than 24 days early, and about 90% submitted it before the deadline. Only about 11% of non-completers ever submitted it.

5. Machine Learning & Results

5.1 OULAD — early-warning classification (O-H1, O-H3) and trajectory clustering

Setup. We compare L2 logistic regression (standardised, class-balanced) with a Random Forest (300 trees, min leaf 10, balanced class weights; Breiman, 2001). Both use scikit-learn (Pedregosa et al., 2011). Validation is **temporal**: we train on all 2013 presentations and test on all 2014 presentations, so the model always predicts a later cohort. We repeat this at nine cut-offs (weeks 1–26), each with three nested feature sets (Table 1). We also ran 5-fold stratified cross-validation within the 2013 data at week 8 (RF AUC 0.827 $\pm$ 0.006). It is close to the temporal test score, so the model generalises across years.

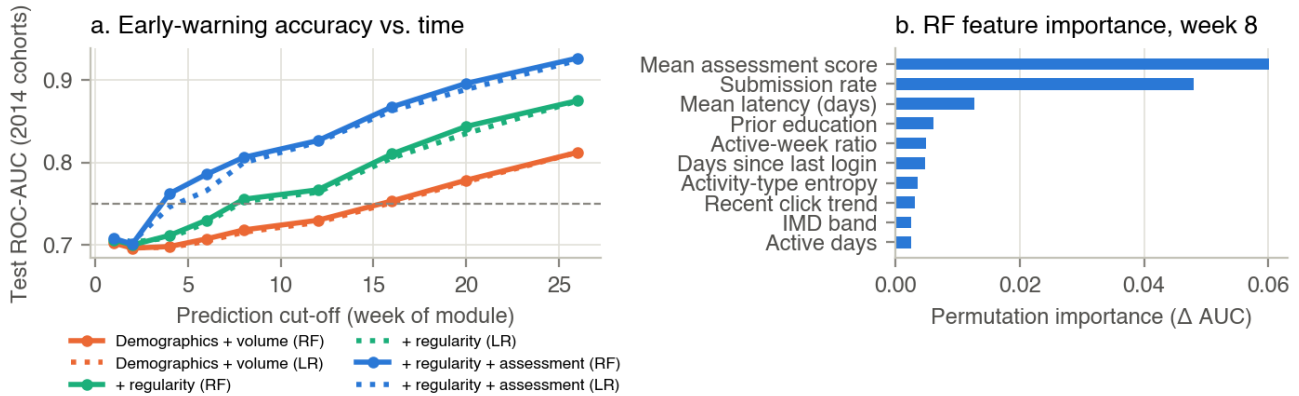


Figure 5. (a) Test ROC-AUC on 2014 cohorts by prediction cut-off and feature set (solid = Random Forest, dotted = logistic regression; dashed line = 0.75 target). (b) Permutation importance (drop in AUC) for the week-8 RF.

Model (week 8, 2014 test, n = 15,098)	ROC-AUC	F1	Precision	Recall	Balanced acc.
Logistic regression (all features)	0.800	0.670	0.685	0.656	0.720
Random Forest (all features)	0.806	0.680	0.673	0.688	0.725
RF, demographics + volume only	0.718				
RF, + regularity (no assessment)	0.755				

Table 2. Week-8 early-warning performance (threshold 0.5; at-risk prevalence among learners still enrolled = 42%).

Results. Prediction accuracy rises steadily with time (Fig. 5a). With demographics and click volume only, the RF reaches AUC 0.70 at week 4 and passes 0.75 only around week 16. Adding regularity features lifts AUC to 0.71 at week 4, 0.76 at week 8 and 0.88 at week 26. Adding assessment behaviour gives 0.76, 0.81 and 0.93. At week 8 the RF identifies 69% of at-risk learners at 67% precision. Logistic regression is nearly as good, which suggests the signal is largely monotonic. The most important week-8 features (Fig. 5b) are mean score, submission rate and latency, followed by prior education, active-week ratio, recency and activity diversity. Total clicks matter much less once regularity is known.

O-H2 test. We fitted a logistic regression of at-risk status on standardised mean latency, mean score and submission rate (n = 25,507 enrolments with ≥ 1 submission). Latency has **no independent effect** (OR per SD = 1.00, 95% CI 0.95–1.05, p = 0.99). Score (OR 0.25) and submission rate (OR 0.06) are both highly significant (p < 0.001).

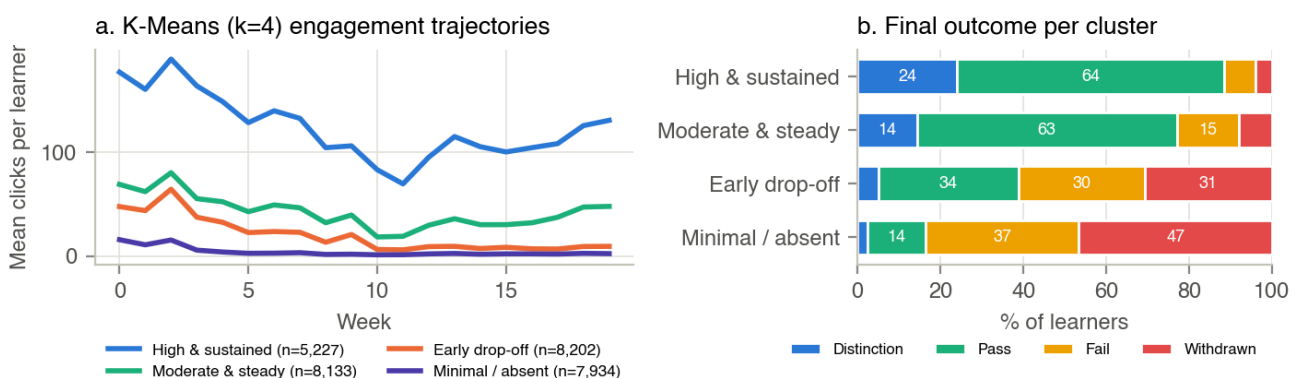


Figure 6. (a) Centroid trajectories from K-Means (k = 4) on log weekly clicks over weeks 0–19. (b) Final-outcome mix within each cluster.

Clustering. We ran K-Means on log-transformed 20-week click trajectories. Silhouette favours $k = 2$ (0.33), a simple engaged/disengaged split. We report $k = 4$ (silhouette 0.18) because it separates the *timing* of disengagement. The four profiles are *high & sustained* (12% at risk), *moderate & steady* (23%), *early drop-off* (61%) and *minimal/absent* (84%) (Fig. 6). The early drop-off cluster starts at a similar level to the moderate cluster but falls away after week 3. These are the learners an intervention could still reach.

5.2 Moodle — completion prediction (M-H1, M-H2) and profile discovery (M-H3)

Setup. With $n = 161$, a single train/test split would be unstable. We therefore use **repeated stratified 5-fold cross-validation (10 repeats, 50 folds)** and report mean $\pm$ SD. The models are an L2 logistic regression ($C = 0.3$) and a Random Forest (400 trees, min leaf 3), both class-balanced. Two baselines put the results in context: logistic regression on action count alone, and an RF without any submission-related or self-monitoring features. Univariate group differences use Mann–Whitney U tests with Holm correction across 19 features.

Model (weeks 0–9 features $\rightarrow$ Final submitted)	ROC-AUC	F1	Precision	Recall	Balanced acc.
Logistic regression (all 19 features)	0.939 $\pm$ 0.045	0.849 $\pm$ 0.062	0.814 $\pm$ 0.079	0.895 $\pm$ 0.084	0.887 $\pm$ 0.050
Random Forest (all 19 features)	0.943 $\pm$ 0.037	0.857 $\pm$ 0.063	0.832 $\pm$ 0.080	0.890 $\pm$ 0.081	0.892 $\pm$ 0.051
RF, behaviour only (no Synopsis / self-monitoring)	0.739 $\pm$ 0.078				
Baseline LR, action count only	0.725 $\pm$ 0.091				

Table 3. Moodle cross-validated performance (mean $\pm$ SD over 50 folds).

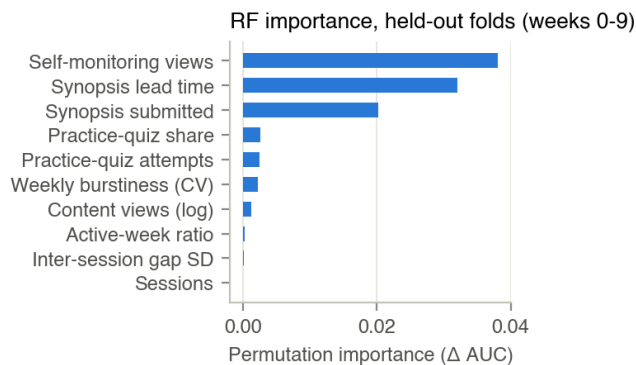


Figure 7. Random-Forest permutation importance, measured on held-out folds. Importance is costly to compute, so it uses 20 folds (5-fold $\times$ 4 repeats) rather than the 50 folds (5-fold $\times$ 10 repeats) behind Table 3.

Feature (weeks 0–9)	Median: completed	Median: not	Rank-biserial r	Holm p
Synopsis lead time (days)	14.34	-5.00	+0.84	< 0.001
Self-monitoring views	16.00	0.00	+0.90	< 0.001
Active days	14.00	7.00	+0.53	< 0.001
Inter-session gap SD (days)	3.24	6.31	-0.51	< 0.001
Active-week ratio	0.70	0.50	+0.50	< 0.001
Sessions	17.00	8.00	+0.49	< 0.001
Time on task (h)	3.10	1.08	+0.43	< 0.001
Practice-quiz share	0.00	0.00	+0.05	1.00
SCORM self-check score	40.58	40.58	-0.00	1.00

Table 4. Selected univariate comparisons between completers ($n = 58$) and non-completers ($n = 103$). Note: learners who never submitted the Synopsis are assigned a lead time of -5 days. The non-completer median of -5.00 is therefore this imputed value (most non-completers never submitted), not an observed lead time.

Results. Both models predict Final completion from week-0–9 behaviour with $AUC \approx 0.94$ (Table 3). However, Fig. 7 and the behaviour-only ablation show that the signal is concentrated in the Synopsis features (whether and how early it was submitted) and in self-monitoring. Removing them drops AUC to 0.74, barely above the action-count baseline (0.73). The univariate tests (Table 4) still support M-H1. Completers were active on twice as many days, had more regular gaps between sessions ($r = -0.51$) and checked their grades and feedback far more often ($r = +0.90$). M-H2 is split. We grouped Synopsis timing into four categories and counted Final completers in each: not submitted, 5 of 96; less than 1 day before the deadline, 1 of 3; 1–7 days before, 11 of 15; more than 7 days before, 41 of 47. The 4×2 table gives $\chi^2(3) = 102$, $p < 0.001$. Practice-quiz use, by contrast, did not differ between groups (Holm $p = 1.00$).

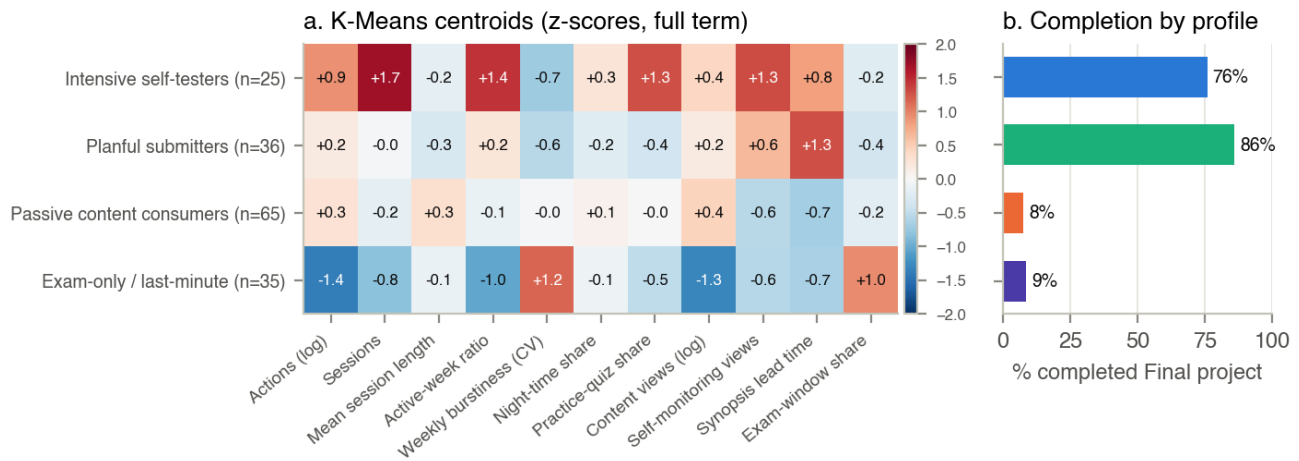


Figure 8. (a) K-Means ($k = 4$) centroids on 11 standardised whole-term features (red = above cohort mean). (b) Share of each profile that completed the Final project. The Synopsis lead-time column includes the -5 -day value assigned to non-submitters. Features cover the whole term, including the period after the outcome, so the profiles describe behaviour and do not predict it.

Profiles (M-H3). K-Means on 11 standardised whole-term features (Fig. 8) produced four interpretable groups. *Intensive self-testers* ($n = 25$) have the most sessions, the most active weeks and the heaviest practice-quiz use (76% completed). *Planful submitters* ($n = 36$) are only moderately active but submit very early (17 days ahead) and check feedback (86% completed). *Passive content consumers* ($n = 65$) read as much material as anyone, in longer sessions, but rarely submit or self-monitor; only 8% completed. *Exam-only / last-minute* learners ($n = 35$) put 46% of all their activity into the two CE weeks; only 9% completed the Final. Completion differs strongly across clusters ($\chi^2 p < 0.001$). Silhouette values are low, however (between 0.18 and 0.21 for every k from 2 to 7). The profiles are best read as regions of a continuum rather than sharply separated types.

6. Discussion & Conclusion

Hypothesis	Verdict	Key evidence
O-H1 early divergence	Supported (partly)	Trajectories separate from week 0 ($r = 0.30$, rising to 0.47 by week 6). Without assessment data (demographics + clickstream), AUC at week 4 = 0.71 (< 0.75); 0.75 is reached by week 8 (0.76), or by week 4 once assessment data are added (0.76).
O-H2 latency	Not supported (independently)	Strong bivariate gradient (25% $\rightarrow$ 76% at risk), but OR = 1.00 after controlling for score and submission rate.
O-H3 regularity	Supported	+0.04 AUC at week 8 and +0.06 at week 16 over volume. Active-week ratio and recency outrank total clicks.
M-H1 distributed / self-monitoring	Supported	Active days, session regularity and self-monitoring all differ (Holm $p < 0.001$). Self-monitoring is the top predictor.
M-H2 procrastination / self-testing	Partly supported	Early Synopsis submission strongly predicts completion and non-submission predicts failure to complete ($\chi^2(3) p < 0.001$). Lateness cannot be tested (inferred deadline). Practice-quiz use does not differ ($p = 1.00$).
M-H3 profiles	Supported (weak structure)	Four interpretable profiles; completion ranges from 8% to 86%. Silhouette ≈ 0.18 .

Table 5. Summary of hypothesis tests.

Interpretation. The two datasets tell the same story. *When* and *how regularly* learners engage reveals more than how much they click, and the most informative behaviours involve meeting early milestones and monitoring one's own progress. These are the behaviours that self-regulated learning theory points to. The OULAD latency result is a useful caution. Late submission looks like a risk factor, but it adds nothing once the model knows whether the learner submits at all and how well they score. Lateness appears to be a symptom of the same underlying disengagement, not a separate cause.

Implications for course design and automated intervention. (1) *Intervene early, and use assessment data.* An OULAD-style alert at week 4 that combines regularity with the first assessment already reaches AUC ≈ 0.76 . Making the first assessment small and early therefore gives the alert system data as well as giving learners practice. (2) *Monitor regularity, not volume.* Dashboards should show active weeks and days since the last login. A learner with many clicks crammed into one week is not on track. (3) *Treat the first milestone as a sensor.* In Moodle, not submitting the Synopsis identified 91 of 103 eventual non-completers by 30 Nov, while flagging only 5 completers, four weeks before the Final opened. A deadline-plus-48-hours nudge is a cheap, explainable trigger. (4) *Prompt self-monitoring.* Completers checked feedback and grade reports far more often. Pushing feedback to learners and adding progress widgets may help the passive and exam-only groups adopt the habit. Whether this would change outcomes is causal and should be tested with an A/B design.

Limitations. Our analyses are correlational. The Moodle outcome is project completion, not a grade, and the cohort is a single course of 161 learners. Synopsis and self-monitoring features are close in kind to the outcome itself, and the Synopsis deadline is inferred rather than taken from the course calendar. Time-on-task depends on the 30-minute session threshold. OULAD click counts do not record dwell time. Future work could add sequence models (e.g. HMMs over weekly states), calibrate the alert thresholds to intervention capacity, and audit fairness across IMD and disability groups before deployment.

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Appendix — code. All analysis is reproducible from https://github.com/hariharateja/ET610_ASSIGNMENT2: code/oulad_analysis.py (OULAD pipeline, ≈ 1 min), code/moodle_analysis.py (Moodle pipeline); intermediate results are in outputs/.